Reconstructing the methane surge with a 2019 reference budget

Reproducible research report

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Abstract

Methane growth reflects both changes in sources and sinks and continued adjustment toward an existing equilibrium. We implement the annual budget decomposition of He et al. in Python and examine the publicly accessible 2019–2022 results without performing a flux inversion. Digitized central curves reconstruct their first two figures over the requested period; published tables and archived fluxes support additional comparisons. The 2020 emission anomaly ranges from about 7 to 37 Tg CH_4 per year among the available reconstructions. Using NOAA annual surface methane as a common burden proxy, the additional removal caused by rising methane amounts to 3.4, 7.8, and 12.3 Tg yr^{-1} in 2020–2022 for a 10-year chemical lifetime. These corrections do not remove differences in the studies' 2019 reference states or OH diagnostics. The archived Peng fluxes reproduce the published 2020 anomaly to within 10^{-6} Tg when integrated over calendar years; normalizing both years to 365 days changes that anomaly by 1.53 Tg. All nine listed papers and their supporting materials were reviewed. Qu (2022) supplies a native burden-based OH forcing of 4.66 Tg yr^{-1} from rounded table entries. Ciais's headline OH fraction uses a different temporal operator and cannot be compared directly with He's forcing fraction. Exact posterior reproduction remains limited by the numerical output available.

1 Scope and available evidence

The target is the steady-state adjustment of He et al. [2026], not a new estimate of methane fluxes from atmospheric observations. We use 2019 as the reference in every anomaly calculation and retain 2019–2022. Absolute baseline quantities appear only where needed to interpret the anomalies. All nine bibliography entries were read in full: He et al. [2026], Ciais et al. [2026], Feng et al. [2023], Laughner et al. [2021], Peng et al. [2022], Qu et al. [2022, 2024], Stevenson et al. [2022], Lawrence et al. [2001]. This includes their supplementary PDFs, appendices, and numerical workbooks. Standalone one-page reading notes are retained in `literature/`. Publisher access failures were resolved with locally supplied Ciais and Qu (2022) articles and supplements, and Laughner's four supplementary files. Their identities, page counts, and checksums are recorded in `data/provided_sources.json`. The original PDFs are retained locally; extracted numerical inputs and all project code, notes, figures, and report sources are in the repository.

2 Budget decomposition and scientific decisions

Let m_j denote annual representative atmospheric methane mass, E_j gross emissions, L_j positive soil uptake, and $k_j = k_{\mathrm{OH,j}} + k_{\mathrm{other,j}}$ the chemical loss frequency. Mass is in Tg CH_4, fluxes in Tg CH_4 yr^{-1}, and loss frequencies in yr^{-1}. The annualized global budget is

$$G_j = E_j - k_j m_j - L_j, \tag{1}$$

in which G_j is the atmospheric growth rate. A subscript 0 denotes 2019 and $\Delta x_j = x_j - x_0$ within each reconstruction. Expanding gives

$$G_j = \underbrace{G_0 - k_0 \Delta m_j}_{R_j} + \underbrace{\Delta E_j}_{F_{E,j}} \underbrace{- m_0 \Delta k_{\mathrm{OH,j}}}_{F_{\mathrm{OH},j}} + \epsilon_j, \tag{2}$$

$$\epsilon_j = -m_0 \Delta k_{\mathrm{other,j}} - \Delta k_j \Delta m_j - \Delta L_j. \tag{3}$$

He neglects ϵ_j. The code returns both this approximation and the exact residual, allowing its validity to be checked when all inputs are available. Soil uptake is not included a second time in k_0. An increasing burden makes the relaxation contribution R_j smaller even when emissions and chemical loss frequencies are unchanged. In anomaly form,

$$\Delta G_j + k_0 \Delta m_j \simeq F_{E,j} + F_{\mathrm{OH},j}. \tag{4}$$

The adjustment has a positive sign on the left: growing methane increases removal under the reference loss frequency. A growth-rate anomaly by itself understates the forcing needed to sustain that growth.

Counterfactual definition. For constant E_0, L_0, k_0 and initial burden $m_c(0) = m_0$, the separately integrated counterfactual is

$$m_c(t) = m_0 + (E_0 - L_0 - k_0 m_0) \frac{1 - e^{-k_0 t}}{k_0}. \tag{5}$$

This is an exact constant-parameter box-model solution. It evaluates losses on $m_c(t)$; Eq. 2 evaluates reference losses on the actual burden m_j. These are distinct diagnostics when the actual and counterfactual trajectories diverge. We implement both and do not equate their integrals. The dashed curves in the reconstructed He figure are digitized published curves, because the native initial burden and all reference losses are not provided numerically in the acquired materials.

Hydroxyl radical (OH) sink definition. If a study supplies positive OH removal $Q_j = k_{\mathrm{OH,j}} m_j$, the equivalent baseline-burden forcing is

$$F_{\mathrm{OH},j} = Q_0 - Q_j \frac{m_0}{m_j}. \tag{6}$$

It is generally different from the raw removal saving $Q_0 - Q_j$. An area- or mass-weighted mean OH change cannot simply replace Δk_{OH}. Lawrence et al. [2001] show that averaging weights, temperature, and the atmospheric domain change the relationship between mean OH and oxidation. We therefore do not convert Feng's latitude-band OH diagnostic or Qu's area-weighted supplementary OH curve directly to a flux. Nor do we substitute a methane perturbation lifetime, which incorporates chemical feedback, for He's chemical lifetime.

Common-burden sensitivity. For Qu (2024)'s tabulated OH removal, a native application of Eq. 6 requires unavailable matched annual model burdens. We additionally evaluate it with $m_j^* = 2.77 X_j$, in which X_j is NOAA's global annual surface mean in ppb and 2.77 Tg ppb^{-1} follows He. This assumption is explicit: a surface mean is a proxy for the whole-atmosphere burden. Its fractional growth, rather than its constant multiplicative conversion, controls Eq. 6. The result is labeled "common burden," not native Qu forcing. Qu (2022), in contrast, tabulates its native burdens and does not require this substitution. To evaluate Eq. 4 for NOAA growth, we examine chemical lifetimes of 9, 10, 11, and 12 years. This sensitivity range brackets the roughly 11-year OH lifetimes reported by He and Qu while allowing additional chemical losses; it is not a fitted or independently inferred lifetime. Soil uptake and other chemical frequencies are held fixed in this diagnostic.

Signed removal changes and prescribed OH scenarios. Ciais Fig. S3 supplies signed OH sink $S = -Q$ relative to 2010, but not absolute 2019 removal or annual model burden. We subtract each series' 2019 value, $s_j = S_j - S_0$, so every plotted anomaly uses 2019. For an assumed OH-only lifetime $\tau_{\mathrm{OH},0}$ and NOAA burden proxy,

$$F^*_{\mathrm{OH},j} = \frac{m^*_0}{m^*_j}\left(s_j + \frac{m^*_j - m^*_0}{\tau_{\mathrm{OH},0}}\right). \tag{7}$$

We use 11.1 yr as a benchmark, motivated by He's reported mean OH lifetime, and test 9–12 yr. This does not establish Ciais's native 2019 lifetime. The same benchmark translates a uniform prescribed OH reduction r into

$$F^*_{\mathrm{OH},2020} = r\, m^*_0/\tau_{\mathrm{OH},0}. \tag{8}$$

These calculations hold baseline weighting and temperature fixed and are conditional comparisons, not recovered posterior results. The derivation is retained in `docs/derivations/published_removal_anomalies.tex`.

Time and uncertainty. NOAA reports annual mean concentrations and growth from January to January. These are different observables and are retained with their own labels. The downloaded NOAA files were created in September 2026 and are not the November 2025 snapshot cited by He. Annual mean burden approximates the burden relevant to an annual budget; no subannual covariance is reconstructed. The original NOAA annual-mean and growth files are archived with this project; their public endpoints are annual means and annual growth. Missing years and missing sink components remain missing. We do not combine annual posterior standard deviations into anomaly errors without their cross-year covariance, and we do not invent common uncertainty intervals for studies with different ensembles and uncertainty definitions.

3 Reconstructing the He figures

Figure 1 reconstructs the 2019–2022 portion of He's Fig. 1 from the public raster and adds the current numerical NOAA record. The plotted quantity is concentration change, despite the growth-rate wording in the original title. Figure 2 reconstructs the central curves of all four panels in He's Fig. 2, expressing emissions, lifetime, sectors, and growth as anomalies from 2019. The original uncertainty envelopes are not reproduced. Panel B subtracts the 2019 relaxation term, so the relaxation anomaly is negative while the original absolute relaxation contribution remains positive.

The file `data/he_digitization.json` records the selected pixels and axis calibrations. A two-pixel coordinate allowance corresponds to about 1.1 Tg yr^{-1} for an individual total emission value and 0.5 Tg yr^{-1} for an individual stacked-bar boundary. Differences between two selected points can have twice those worst-case bounds. These are digitization tolerances, not posterior uncertainty. The independently digitized emission changes in panels A and B agree to within 0.35 Tg yr^{-1}. This checks transcription consistency, not the validity of the inversion.

4 Comparing emission and sink changes

Table 1 and Fig. 3 show the available annual global anomalies. The largest emission disagreement is in 2020: Qu (2024)'s 37 Tg yr^{-1} exceeds digitized He by about 29 Tg yr^{-1} and Feng's joint-OH estimate by 19.1 Tg yr^{-1}. The He–Qu difference narrows to about 9 Tg yr^{-1} in 2021, then widens to about 16 Tg yr^{-1} in 2022. Qu (2022) gives 31 Tg yr^{-1} in 2020, while Ciais reports 22.0 ± 3.5 Tg yr^{-1}. The digitized Ciais central emission changes are 21.9, 24.9, and 23.4 Tg yr^{-1} in 2020–2022. Thus Ciais and He also

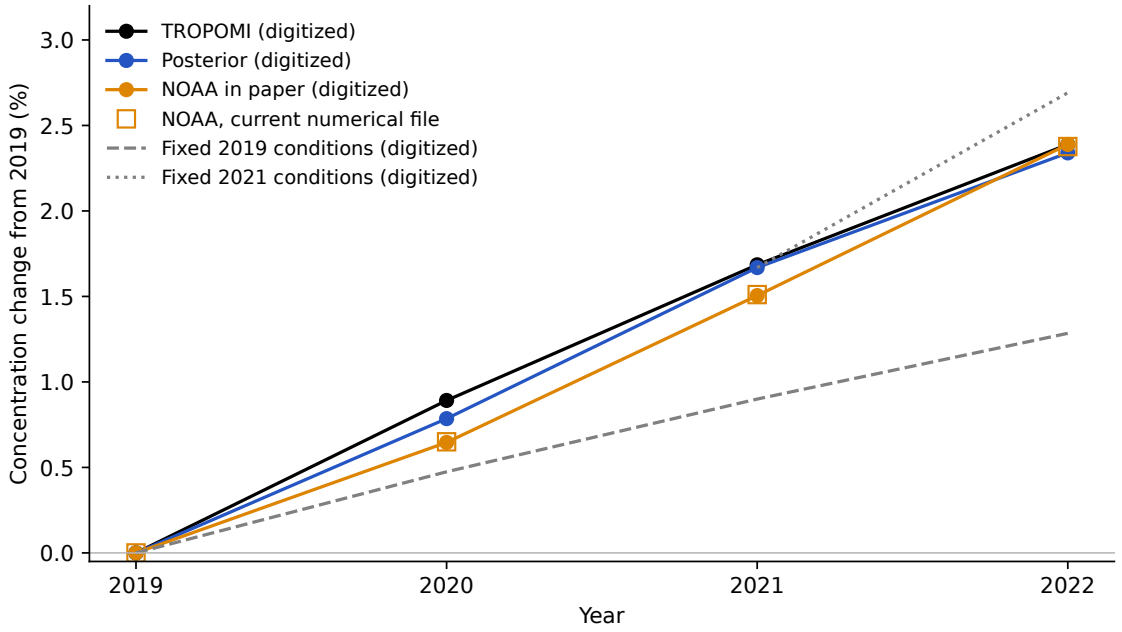


Figure 1: Reconstruction of He Fig. 1 for 2019–2022. Black, blue, and orange filled markers are digitized TROPOMI, posterior GEOS–Chem, and published NOAA concentration changes, respectively. Orange open squares use the September 2026 NOAA numerical file. Gray dashed and dotted curves are the published fixed-2019 and fixed-2021 counterfactuals, respectively, digitized without refitting. All concentration changes are expressed as percentages of their own 2019 values. The fixed-2021 branch retains the same 2019 reference.

differ by about 21 Tg yr^{-1} in 2022. These comparisons preserve each study's own 2019 emissions. They do not imply that the reconstructions share the same absolute emission baseline.

The sink correction grows through 2022. For Qu (2024), raw OH removal savings of 11, 10, and 25 Tg yr^{-1} become about 14, 17, and 36 Tg yr^{-1} under the common-burden calculation. This occurs because maintaining even the reduced removal flux at a larger burden requires a larger decline in loss frequency than the raw flux change suggests. At a 10-year chemical lifetime, the NOAA burden adjustment is 3.36, 7.81, and 12.29 Tg yr^{-1}. Varying the lifetime from 9 to 12 years changes the 2022 correction from 13.65 to 10.24 Tg yr^{-1}. The correction therefore matters increasingly after the initial surge, but it does not uniquely identify emissions versus OH.

The baseline is a separate discrepancy. Qu (2024)'s Table S1 reports 2019 emissions of 550 Tg yr^{-1} and total removal of 553 Tg yr^{-1}, implying growth of -3 Tg yr^{-1}. The corresponding NOAA growth in the current file is $+26.70$ Tg yr^{-1}; digitized He is about $+27.7$ Tg yr^{-1}. Qu's 2020 native growth of 45 Tg yr^{-1} thus represents a 48 Tg yr^{-1} anomaly, even though its absolute 2020 growth is near NOAA's 40.94 Tg yr^{-1}. Replacing Qu's baseline with NOAA would change the question and obscure this initialization issue. Qu adjusts its initial concentration each year, while its continuous diagnostic run does not impose those resets. He propagates the posterior state with a Kalman approach. These documented choices are plausible contributors to the baseline difference; the available global tables cannot isolate their individual effects. Qu (2022) gives growth of 28 and 59 Tg yr^{-1}, a 31 Tg yr^{-1} anomaly, versus 14.24 Tg yr^{-1} in our NOAA record. Its annual initial states are independently scaled by 0.97 and 1.01. This native budget difference must be retained even though the initial burden and 2019 emissions are close to He's. It is not corrected by replacing the tabulated model growth with NOAA.

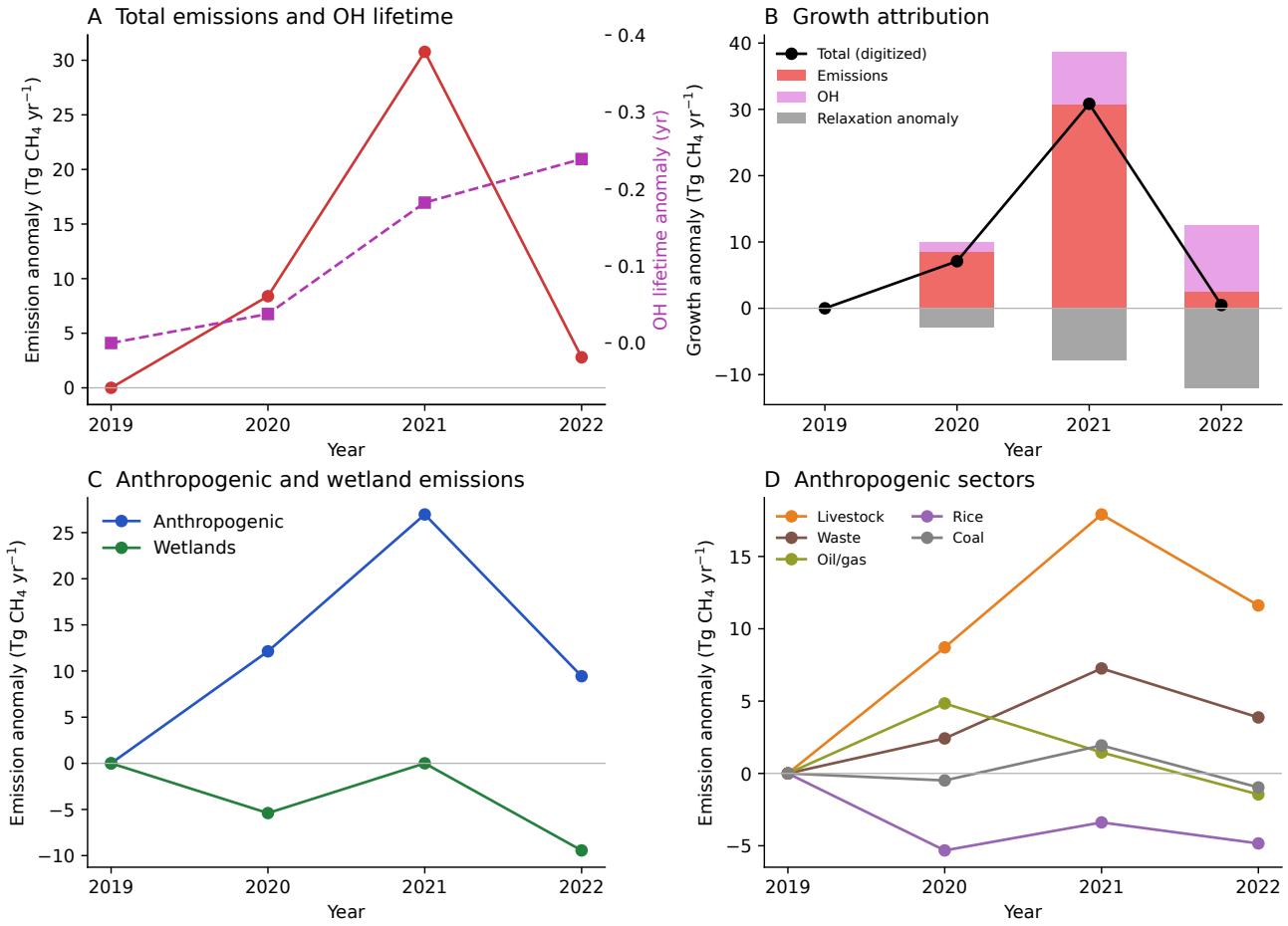


Figure 2: Reconstruction of He Fig. 2 central values, rebased to 2019. (A) Red emission changes and purple changes in lifetime against tropospheric OH. (B) Red emission forcing, pink OH forcing, and gray relaxation anomalies; the black curve is their sum. The original positive baseline relaxation is subtracted. (C) Anthropogenic and wetland emission changes. (D) Changes in the five displayed anthropogenic sectors. Curves are digitized; shaded posterior ranges are omitted. Sector attribution inherits He's prior sector fractions. The panels are not independent flux inversions.

Attribution fractions have different denominators. He reports 81% emissions and 19% OH for the additional forcing during 2020–2021; its 59/25/16% relaxation/emissions/OH partition describes 2019–2024 total growth and is outside this report's period. Peng's 47/53% split concerns the 2020 growth-rate increase. Feng's 34% OH contribution in 2020 is inferred from the reduction in required emissions when OH is optimized: the fixed-OH anomaly is 27.0 Tg yr^{-1} and the joint-OH anomaly is 17.9 Tg yr^{-1}. The 9.1 Tg yr^{-1} difference is a useful sensitivity result but is not itself a separately measured $-m_0\Delta k$. Stevenson's 2.86 ± 1.13 ppb yr^{-1} is a one-year response to precursor changes, equivalent to 7.92 ± 3.13 Tg over a year using He's conversion. We do not average these fractions into a consensus attribution.

Ciais's OH fraction answers a different question. The supplement of Ciais et al. [2026] defines its headline contribution as

$$\bar{f}_{\mathrm{OH}} = \frac{1}{4} \sum_{j=2020}^{2023} \frac{S_j - S_{j-1}}{G_j - G_{j-1}}, \qquad S_j = -Q_j. \tag{9}$$

Study / quantity	2020	2021	2022
He emissions (digitized)	8.4	30.8	2.8
He OH forcing (digitized)	1.5	7.8	10.0
He relaxation anomaly (digitized)	-2.9	-7.8	-12.0
Qu 2022 emissions	31.0	—	—
Qu 2022 OH forcing, native burden	4.7	—	—
Ciais emissions (digitized)	21.9	24.9	23.4
Ciais OH-1, conditional	6.4	22.8	14.1
Ciais OH-2, conditional	7.3	15.3	10.1
Qu 2024 emissions	37.0	40.0	19.0
Qu 2024 OH removal saving	11.0	10.0	25.0
Qu 2024 OH forcing, common burden	14.0	17.0	35.5
Feng emissions, joint OH	17.9	18.4	—
Feng emissions, fixed OH	27.0	20.8	—
NOAA growth anomaly	14.2	22.3	9.3
Burden adjustment, $\tau_0 = 10$ yr	3.4	7.8	12.3

Table 1: Annual anomalies relative to 2019 in Tg CH_4 yr^{-1}. He and Ciais values are digitized central estimates. Qu (2022) uses rounded Table 1; Qu (2024) uses Table S1; Feng uses Table 1. Qu (2022)'s OH forcing uses native burden. Qu (2024)'s common-burden OH forcing uses Eq. 6 and NOAA annual surface means. It is conditional, whereas the raw removal saving is a direct table difference. Ciais OH-1 and OH-2 use Eq. 7 with an assumed 11.1-yr 2019 OH lifetime. The last row is a sensitivity calculation, not another inversion. Dashes denote unavailable years. All anomalies are zero in 2019 by definition.

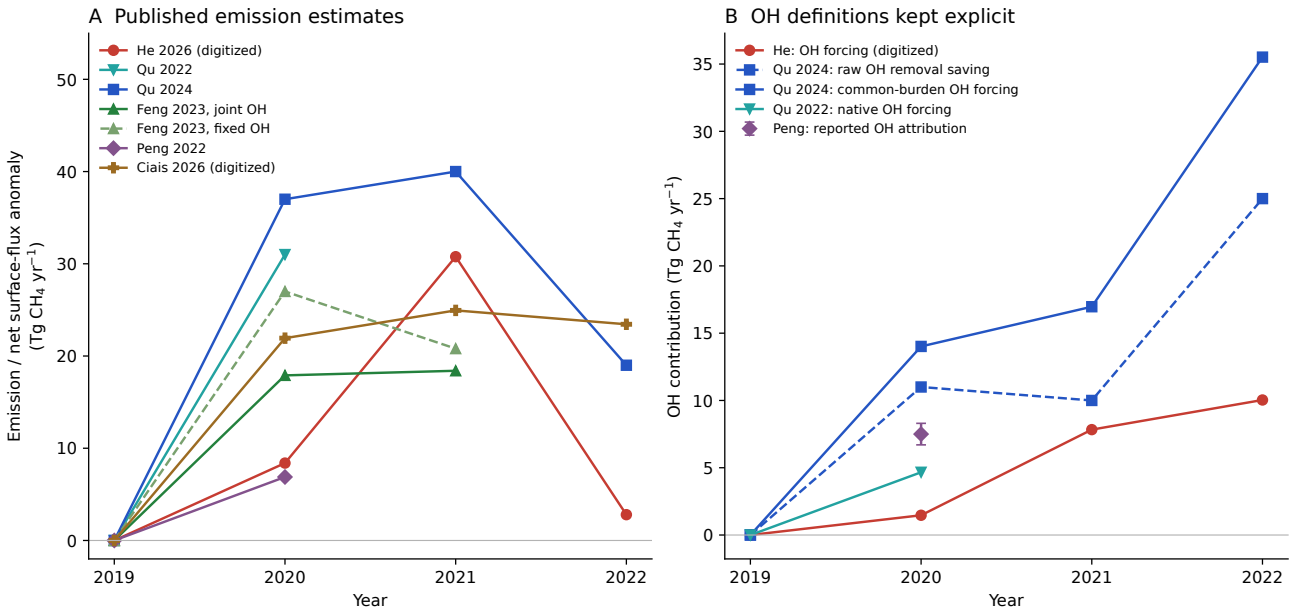


Figure 3: Global emission and OH comparisons. (A) Emission anomalies from He (red), Qu 2024 (blue), Qu 2022 (teal), Ciais (brown), Feng joint and fixed OH (green solid and dashed), and Peng (purple). Peng's calendar-year net surface-flux difference is retained; the time-normalization distinction is quantified below. (B) He OH forcing (red), Qu 2022 native forcing (teal), Qu 2024 raw OH removal saving (blue dashed), and Qu 2024 common-burden forcing (blue solid). The purple diamond and error bar are Peng's reported 7.5 ± 0.8 Tg yr^{-1} OH attribution; its definition is not silently replaced by the Qu diagnostic. Lines end where each study's available period ends.

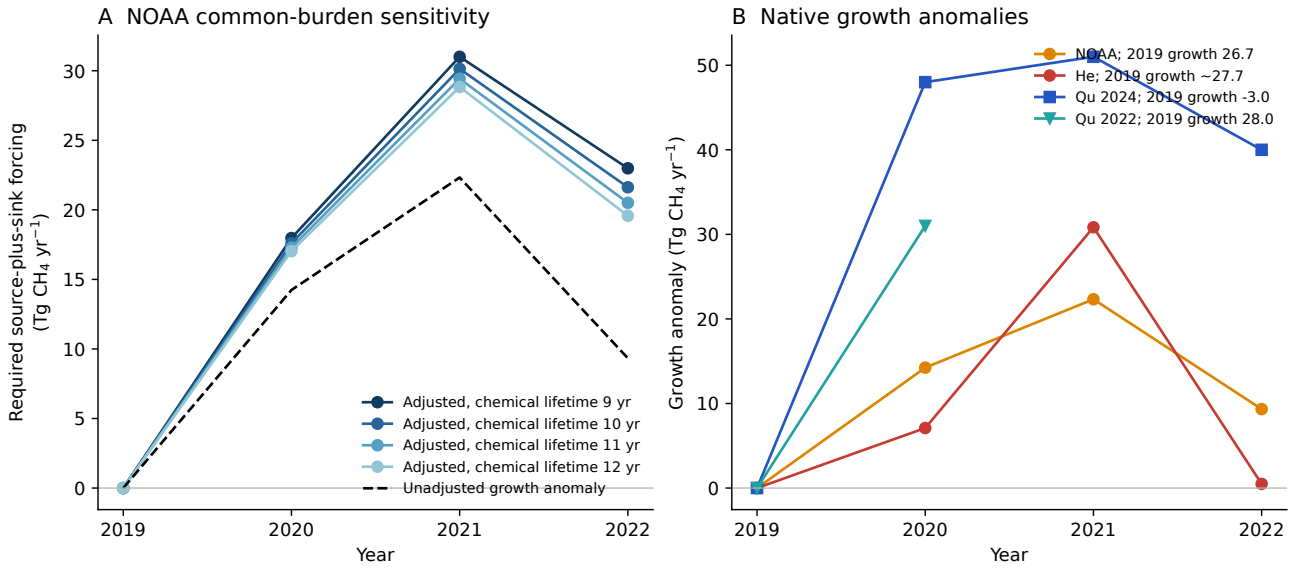


Figure 4: Adjustment sensitivity and native growth anomalies. (A) Blue curves add $\Delta m/\tau_0$ to the current NOAA growth anomaly for chemical lifetimes of 9–12 years; black dashed is unadjusted growth. (B) NOAA (orange), digitized He (red), Qu 2024 (blue), and Qu 2022 (teal), each relative to its own 2019 growth. Legend values state the absolute baseline growth in Tg yr^{-1}. Annual mean surface methane is a burden proxy in A; the curves are not calibrated to reproduce the posterior estimates in B.

It averages fractions of adjacent-year growth changes, rather than ratios of fixed-2019 source and OH forcings. The denominator can be small or change sign, and 2023 lies outside the requested interval. The abstract reports 83%; the main discussion gives 70–86% for NOAA, whereas the conclusion gives 74–80%. We retain those distinctions and do not reinterpret the headline percentage as variance explained or as $F_{\mathrm{OH}}/(F_E + F_{\mathrm{OH}})$. The separately reported regression R^2 values are 0.5–0.7.

Figure 5 instead uses annual inversion-mean removal circles from Ciais Fig. S3. Their raw 2019-based savings are approximately 3.4, 16.1, and 3.4 Tg yr^{-1} for OH-1, and 4.4, 8.5, and −0.7 for OH-2. The conditional 11.1-yr correction gives 6.4, 22.8, and 14.1 for OH-1, and 7.3, 15.3, and 10.1 for OH-2. The largest 2022 OH comparison remains Qu (2024)'s 35.5 versus He's digitized 10.0 Tg yr^{-1}, although the former depends on a substituted burden. Native Ciais closure cannot be checked without the matching burden, growth diagnostic, and other losses.

Laughner's scenario is now constrained by its supplement. Laughner et al. [2021] prescribe a 3% OH decrease during 2020 in a two-hemisphere box model with the same emissions in perturbed and control runs. Their 2–4% range comes from March–May temporal variation in a chemistry-assimilation experiment, not an annual posterior confidence interval. Equation 8 gives a conditional 2020 forcing of 14.0 Tg yr^{-1} for 3% and 11.1 yr, or 9.3–18.6 for 2–4% at the same lifetime. Jointly varying the lifetime over 9–12 yr broadens that range to 8.6–23.0. These are scenario equivalents, not new inferred fluxes. The paper's plotted 2021 trajectories are explicitly extrapolated, not box-model predictions. Its inventory workbook has no methane values for 2018–2020, so it cannot supply a 2019-baseline emission reconstruction.

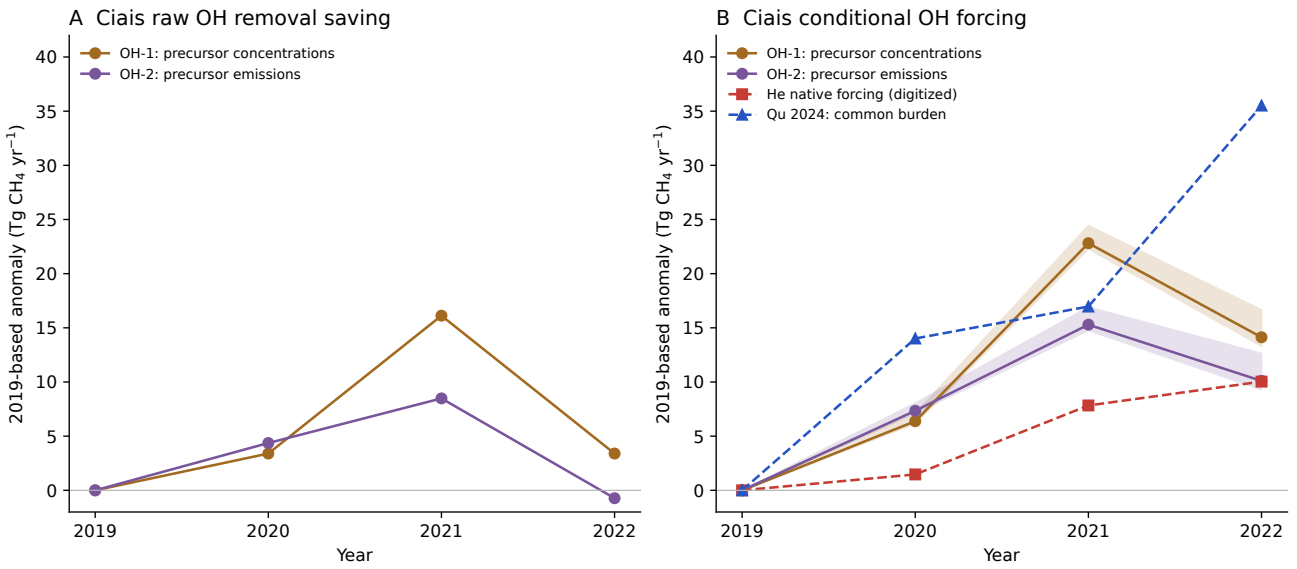


Figure 5: Ciais removal changes and the conditional steady-state adjustment. (A) Brown and purple show OH-1 and OH-2 signed-removal changes after subtracting 2019. (B) The same curves use Eq. 7 with NOAA burden and an assumed 11.1-yr OH lifetime; shading spans 9–12 yr and is not a posterior uncertainty interval. Red dashed is He's digitized native OH forcing; blue dashed is Qu (2024) with NOAA burden. Ciais's two-pixel reading allowance implies a worst-case difference error of about 1 Tg yr^{-1}, separate from both model spread and lifetime sensitivity. The published 2010 reference and the offset right axis are not used as anomaly baselines.

5 Why the reconstructions differ

Tropical wetlands versus livestock. Qu (2024) assigns 43% of its emission surge to Equatorial Asia and 30% to Africa; Feng's fixed-OH inversion gives a 2020 East African increase of 14 ± 3 Tg relative to 2019. He finds a strong 2021 increase but assigns much of its global source change to anthropogenic emissions, with wetlands approximately unchanged from 2019 to 2021. This attribution depends on the prior spatial overlap of livestock and wetlands: He optimizes their total grid-cell flux and divides it using prior sector fractions, while Qu optimizes monthly regional wetland fluxes separately. He's precipitation-driven wetland prior can miss inundation displaced downstream from rainfall. Its independent ammonia comparison does not support increased African livestock emissions. Thus a livestock label in the posterior is not a direct observation of a livestock increase. Ciais applies a different additional assumption: in mixed cells, source changes are reallocated when their agricultural fraction differs by more than 20% from the corresponding bottom-up fraction. Its unmodified prior-proportional allocation would assign $46.8 \pm 14.8\%$ to agriculture and waste and $34.7 \pm 4.6\%$ to wetlands. Its wetland attribution is therefore not a direct observation either. Ciais's air-mass-weighted OH maps and isotope-box source tests constrain different aspects of the budget; neither provides a replacement for annual reaction-weighted k_{OH}.

South America and the baseline interval. Feng finds positive 2020 tropical South American emission changes and larger 2021 tropical and temperate South American increases. Qu (2024) emphasizes that much of its South American increase preceded 2020, so it contributes less to the 2020–2022 surge relative to 2019. He reports 2019–2024 trends, a different endpoint interval that cannot be relabeled as 2019–2022. Without common regional posterior arrays, a difference map or a precise regional ranking across these reconstructions would imply information that is not available. The global 2020 emission and 2022 OH differences are quantified; regional explanations remain method-based comparisons. Qu

Table 2: Methods of all nine reviewed studies. These distinctions affect source attribution, reference states, and sink meaning; they do not establish a ranking or isolate the effect of each modeling choice.

Study	Observations and transport	Optimized or prescribed OH	Source and reference treatment
He 2026	TROPOMI; GEOS–Chem, $2^\circ \times 2.5^\circ$; GEOS-FP	Two annual hemispheric factors; joint inversion	4111 annual land fluxes; preceding posterior as prior; sectors from prior fractions
Qu 2022	GOSAT land and glint; GEOS–Chem, $2^\circ \times 2.5^\circ$; MERRA2	One annual global OH loss-frequency factor; joint inversion	4020 nonwetland and 168 wetland terms; independent annual priors and initial-state scaling
Qu 2024	GOSAT including ocean glint; GEOS–Chem, $4^\circ \times 5^\circ$; MERRA2	Two annual hemispheric factors; joint inversion	Annual fixed prior; 1009 nonwetland grid fluxes and 168 monthly wetland terms; yearly initial-state scaling
Ciais 2026	GOSAT; PYVAR–LMDz, WESTLAKE, and GONGGA, with different resolutions and assimilation windows	Prescribed precursor-concentration (OH-1) or emission-driven (OH-2) chemistry	Flat wetland priors; six model/OH combinations; mixed-cell source changes reallocated using bottom-up fractions
Feng 2023	GOSAT CH_4/CO_2 ratios and surface constraints; GEOS–Chem, $2^\circ \times 2.5^\circ$; MERRA2	Fixed climatology or monthly factors in six latitude bands	Ensemble Kalman filter; 487 regions; four-month lag; regional fluxes
Peng 2022	103 surface sites; LMDZ–SACS, $1.9^\circ \times 3.75^\circ$; ERA5	Prescribed LMDZ–INCA chemistry; independent halocarbon check	Eight-day gridded net flux; variational inversion; bottom-up source attribution
Stevenson 2022	HTAP chemistry response experiments, with 2020 precursor changes	Chemistry sensitivities and perturbation lifetimes	No methane flux inversion; 2001 sensitivities transferred to a 2020 pulse
Laughner 2021	Two TCCON sites; two-hemisphere methane box model	Prescribed 3% reduction during 2020, bracketed by 2–4%	Same stabilized emissions in control and perturbation; observations referenced to a 2016–2019 trend
Lawrence 2001	Four three-dimensional OH fields; global oxidation diagnostics	Compares averaging weights and atmospheric domains	Methodological study; no 2019–2022 reconstruction

(2022) reports a 2020 South American total-emission reduction of 2.7 Tg yr^{-1}. Ciais reports a 2020–2022 mean reduction of 1.4 ± 0.8 Tg yr^{-1} specifically for southern South American wetlands, with a north–south contrast within Amazonia. Those differences in sector, boundary, and averaging interval preclude treating the two numbers as the same regional observable. Across studies, Africa is a more consistent location of increases than South America, but livestock versus inundated wetland attribution remains sensitive to prior structure.

Chemistry and sampling. Peng prescribes three-dimensional OH changes simulated from changes in precursor emissions and meteorology, while He, Qu, and Feng infer low-dimensional OH factors from methane observations. The inferred split depends on prior errors, transport, the spatial sampling of surface versus column observations, and which OH degrees of freedom are allowed. Stevenson shows why a NO_x reduction alone is insufficient: its positive 4.75 ppb yr^{-1} contribution is partly offset by negative CO and NMVOC contributions totaling about 1.90 ppb yr^{-1}. Applying a single OH percentage everywhere would conceal this spatial and chemical structure.

A measured sensitivity of the inversion design. The nine cases in Qu (2022)'s main text and Table S1 give a particularly direct comparison (Fig. 6). Reducing regularization from the base 0.5 to 0.1 shifts the rounded source/OH forcing from 31/5 to 19/20 Tg yr^{-1}; increasing it to 1 gives 34/1. Changing only the OH prior standard deviation from 5% to 20% leaves rounded emission forcing at 31 Tg yr^{-1}. Thus sensitivity cannot be judged from the OH prior alone. The forcing stays within 35–39 across all cases, while the source–sink split varies substantially. These are alternative inversion settings, not independent posterior samples. They demonstrate a mechanism for attribution differences without proving which setting explains the difference from He.

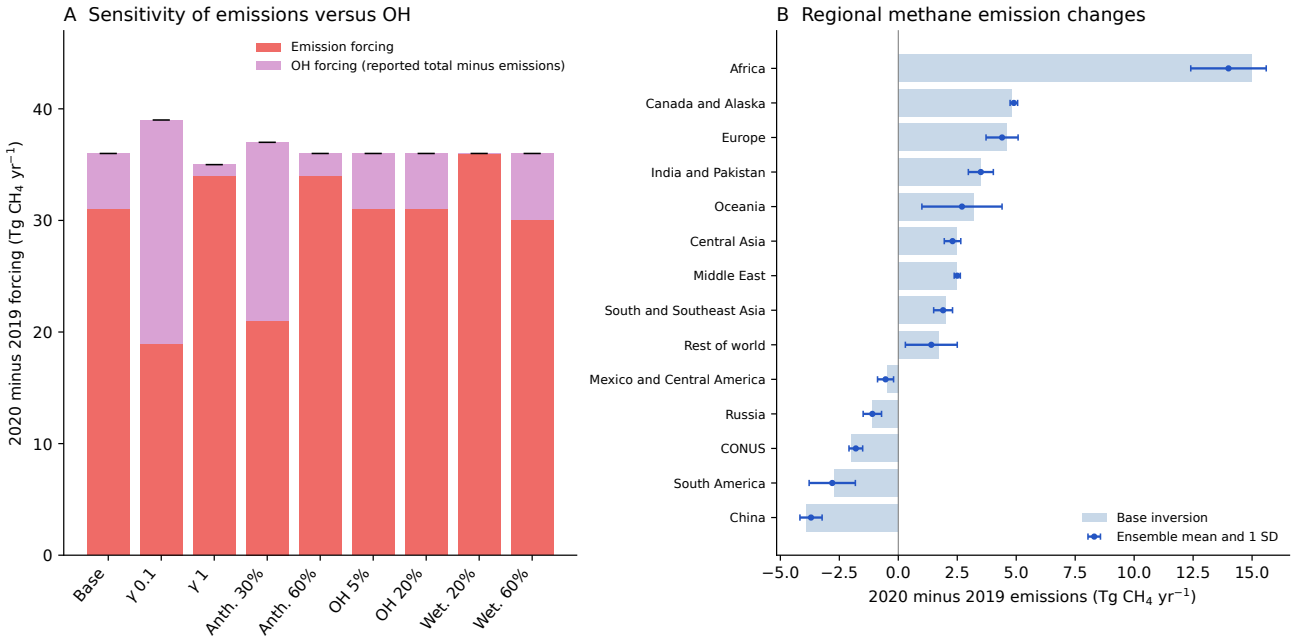


Figure 6: Qu (2022) sensitivity and regional evidence, all comparing 2020 with 2019. (A) Red bars are emission forcing; pink is reported total forcing minus emissions, using rounded main-text and Table S1 values. Black marks give total forcing. Case labels identify regularization or the prior error standard deviation changed from the base case. (B) Light-blue bars are Table 2 base regional emission anomalies; dark-blue points and error bars are the reported ensemble means and one-standard-deviation spreads. These regions do not match Peng's latitude bands and are not silently aggregated into the same mask. The figures are recalculations from published tables, not a repeated inversion.

6 A numerical reproduction of the Peng flux anomaly

Peng's public monthly posterior is supplied in kg CH_4 m^{-2} s^{-1} with grid-cell areas in Figshare record 21076171. We multiply by area and the number of seconds in each calendar month before summing. The reconstructed calendar-year totals are 552.5443 Tg in 2019 and 559.4283 Tg in 2020, giving 6.88403165 Tg. The publisher spreadsheet gives 6.8840322 Tg, agreeing within 10^{-6} Tg. This is a reproduction from numerical output, independent of figure digitization.

The same monthly data give mean rates of 552.5443 and 557.8998 Tg per 365-day year. Their difference is 5.35554 Tg yr^{-1}; the extra 2020 day accounts for 1.52849 Tg of the calendar-total difference. Both quantities are valid when named correctly, but their units and temporal operators differ. We retain the reported calendar total in the comparison and expose the alternative rate normalization. This difference is smaller than the He–Qu emission gap but comparable to some reported source changes. For

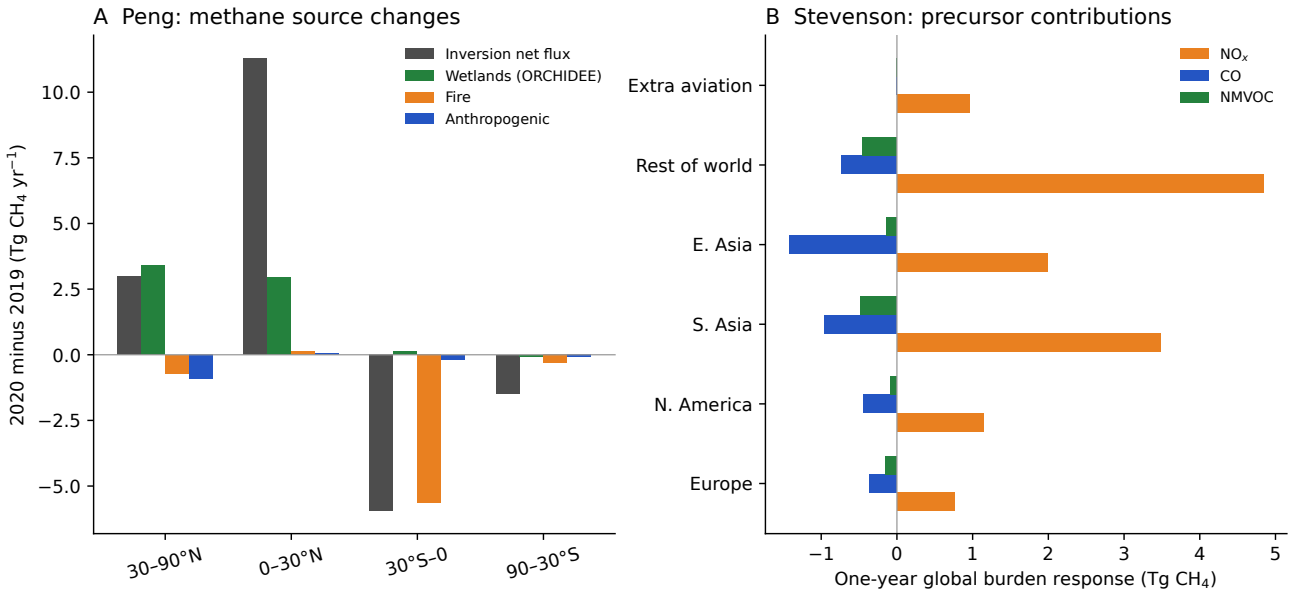


Figure 7: Regional evidence with distinct physical meanings. (A) Peng source spreadsheet: 2020 minus 2019 methane inversion flux (gray), ORCHIDEE wetlands (green), fire (orange), and anthropogenic sources (blue) in four latitude bands. The bottom-up categories need not sum to the inversion. (B) Stevenson Table S13a: global one-year methane burden responses to NO_x (orange), CO (blue), and NMVOC (green) emission changes in each source region, converted with 2.77 Tg ppb^{-1}. Aviation is only the additional correction, avoiding double counting. Panel B is not a map of regional methane emission changes.

the map in Fig. 8, we compare day-weighted annual mean rates, thereby removing the purely calendar-length contribution.

7 Reproducibility and remaining numerical limits

He figure	Reproduction status and required missing information
1	Central 2019–2022 curves digitized; updated NOAA numerical comparison added. Native model mass, complete loss terms, and original NOAA snapshot are needed for an exact independent counterfactual reproduction.
2A–D	Central time series digitized and rebased. Original ensemble members and annual posterior diagnostics are needed for numerical curves and uncertainty envelopes. No posterior inversion was performed.
3A–B	Not reconstructed. The map/pie and regional trends use 2024 or 2019–2024 endpoints and require gridded posterior sector data. They cannot be converted to the target interval from the published image alone.
S1	Fit maps and annual fit statistics require sampled prior/posterior concentrations and observation-error diagnostics.
S2	Averaging kernels and degrees of freedom require posterior covariance and sensitivity matrices.
S3	Sector trend maps require annual gridded sector fluxes; the published 2019–2024 trends cannot determine 2019–2022 anomalies.

He figure	Reproduction status and required missing information
S4	The ammonia trend diagnostic covers 2019–2024 and requires the matched IASI retrieval processing and regional masks. It is not a methane flux series.
S5	The 2024 error-correlation matrix requires posterior covariance.
S6	Prior wetland/anthropogenic overlap requires the actual gridded prior files and masks used by the inversion; model code alone is insufficient.
S7	TROPOMI/Carbon-I information-content experiment requires the specified sensitivity and observation-error matrices; rebuilding it would require work beyond the requested steady-state analysis.
S8	The error-versus-observation-count diagnostic requires individual 2024 residuals and sampling information.
S9	Comparison of regularization choices requires both posterior inversion outputs. Re-running those inversions is explicitly out of scope.

The public He availability statement points to model code and observational inputs, rather than a numerical posterior archive sufficient for the items above. Absence from those acquired materials is not proof that the authors cannot provide the outputs. Complete annual burden, chemical loss, soil, sector flux, and ensemble data would permit a native application of Eq. 2 and much stronger cross-study comparisons. We have not substituted digitized diagnostic matrices or invented spatial flux fields.

The Python analysis, pinned pixi environment, original starter instructions, download manifest with checksums, transcription tables, digitization coordinates, processed netCDF files, executed figure notebook, reading notes, and LaTeX sources are retained in the project repository. Four reusable budget tests check algebraic closure, the constant-rate limit, analytical trajectories, and baseline handling. Native full budgets remain unavailable, so these tests validate the implementation rather than posterior numerical agreement.

All requested literature is now present and reviewed. Qu (2022)'s main and supplementary budget tables, Ciais's bottom-up Table S1 and digitized annual markers, and Laughner's complete workbooks are retained as numerical inputs. The Qu (2022) availability DOI points to GOSAT retrievals; the Ciais statement points to the article and supplement. Neither supplies the full matched posterior budget needed to reproduce all of He's terms for every study. Ciais's other-natural-source treatment is described inconsistently on supplement pp. 4–5 and in the main Fig. 2 caption; no unsupported component is assigned from that ambiguity. Downloading a later inversion product would not make it the posterior used in these papers.

The project reproduces the accessible numerical comparisons and clearly labels graphical extraction and sensitivity calculations. Exact spatial, ensemble, and counterfactual reconstruction of the remaining He figures requires the author's numerical outputs, rather than a new inversion.

References

P. Ciais, Y. Zhu, Y. Cai, X. Lan, S. E. Michel, B. Zheng, Y. Zhao, D. A. Hauglustaine, X. Lin, Y. Zhang, S. Sun, X. Tian, M. Zhao, Y. Wang, J. Chang, X. Dou, Z. Liu, R. Andrew, C. A. Quinn, B. Poulter, Z. Ouyang, W. Yuan, K. Yuan, Q. Zhu, F. Li, N. Pan, H. Tian, X. Yu, G. Rocher-Ros, M. S. Johnson, M. Li, M. Li, D. Feng, P. Raymond, X. Yang, J. G. Canadell, R. B. Jackson, X. Yu, Y. Li, M. Saunois, P. Bousquet, and S. Peng. Why methane surged in the atmosphere during the early 2020s. *Science*, 391(6785):eadx8262, February 2026. doi: 10.1126/science.adx8262.

Liang Feng, Paul I. Palmer, Robert J. Parker, Mark F. Lunt, and Hartmut Bösch. Methane emissions are predominantly responsible for record-breaking atmospheric methane growth rates in 2020 and 2021.

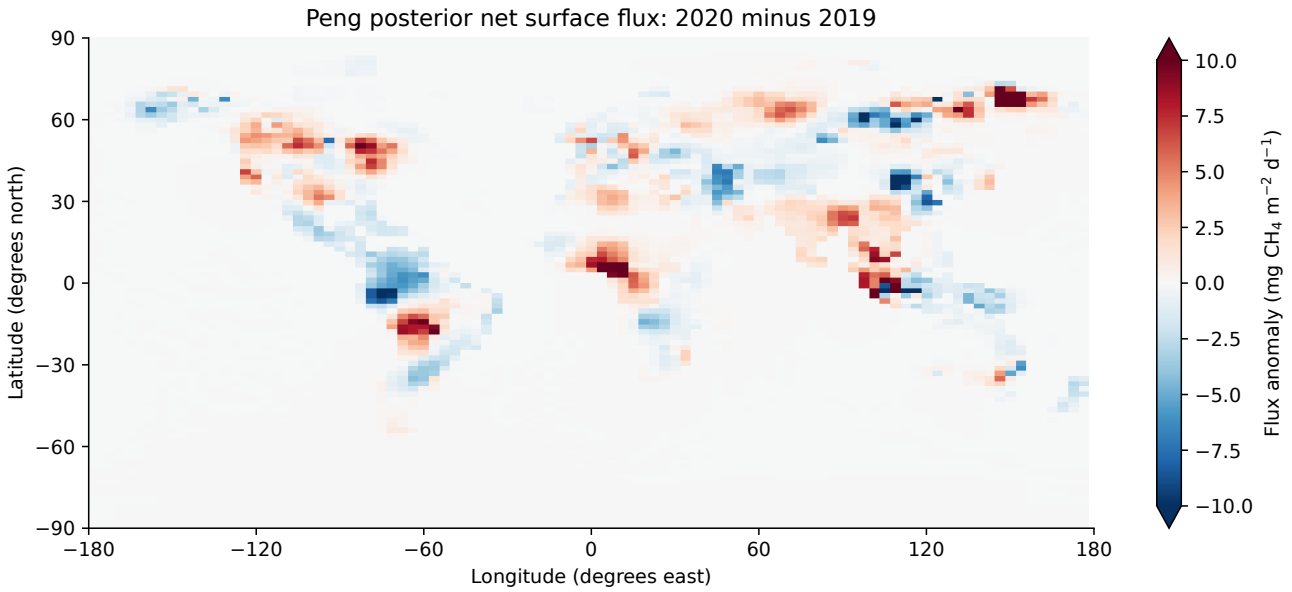


Figure 8: Peng archived posterior net surface-flux rate, 2020 minus 2019. Each annual field is weighted by days per month; colors show mg CH_4 m^{-2} d^{-1} with saturation at ± 10. Coordinates and grid-cell areas come from the archived netCDF. Negative values are reductions relative to 2019. This is a map of net surface flux, including prescribed soil uptake, not a sector-resolved or regional OH attribution.

Atmospheric Chemistry and Physics, 23(8):4863–4880, April 2023. ISSN 1680-7316. doi: 10.5194/acp-23-4863-2023.

Megan He, Daniel J. Jacob, Lucas A. Estrada, Daniel J. Varon, Melissa Sulprizio, Nicholas Balasus, James D. East, Elise Penn, Drew C. Pendergrass, Zichong Chen, Todd A. Mooring, Joannes D. Maasakkers, Philip G. Brodrick, Christian Frankenberg, Kevin W. Bowman, and Lori Bruhwiler. Attributing 2019–2024 methane growth using TROPOMI satellite observations. *Science Advances*, 12 (15):eadz9007, April 2026. doi: 10.1126/sciadv.adz9007.

Joshua L. Laughner, Jessica L. Neu, David Schimel, Paul O. Wennberg, Kelley Barsanti, Kevin W. Bowman, Abhishek Chatterjee, Bart E. Croes, Helen L. Fitzmaurice, Daven K. Henze, Jinsol Kim, Eric A. Kort, Zhu Liu, Kazuyuki Miyazaki, Alexander J. Turner, Susan Anenberg, Jeremy Avise, Hansen Cao, David Crisp, Joost de Gouw, Annmarie Eldering, John C. Fyfe, Daniel L. Goldberg, Kevin R. Gurney, Sina Hasheminassab, Francesca Hopkins, Cesunica E. Ivey, Dylan B. A. Jones, Junjie Liu, Nicole S. Lovenduski, Randall V. Martin, Galen A. McKinley, Lesley Ott, Benjamin Poulter, Muye Ru, Stanley P. Sander, Neil Swart, Yuk L. Yung, and Zhao-Cheng Zeng. Societal shifts due to COVID-19 reveal large-scale complexities and feedbacks between atmospheric chemistry and climate change. *Proceedings of the National Academy of Sciences*, 118(46):e2109481118, November 2021. doi: 10.1073/pnas.2109481118.

M. G. Lawrence, P. Jöckel, and R. von Kuhlmann. What does the global mean OH concentration tell us? *Atmospheric Chemistry and Physics*, 1(1):37–49, December 2001. doi: 10.5194/acp-1-37-2001.

Shushi Peng, Xin Lin, Rona L. Thompson, Yi Xi, Gang Liu, Didier Hauglustaine, Xin Lan, Benjamin Poulter, Michel Ramonet, Marielle Saunois, Yi Yin, Zhen Zhang, Bo Zheng, and Philippe Ciais. Wetland emission and atmospheric sink changes explain methane growth in 2020. *Nature*, 612(7940): 477–482, December 2022. ISSN 1476-4687. doi: 10.1038/s41586-022-05447-w.

Zhen Qu, Daniel J Jacob, Yuzhong Zhang, Lu Shen, Daniel J Varon, Xiao Lu, Tia Scarpelli, Anthony Bloom, John Worden, and Robert J Parker. Attribution of the 2020 surge in atmospheric methane by inverse analysis of GOSAT observations. *Environmental Research Letters*, 17(9):094003, September 2022. ISSN 1748-9326. doi: 10.1088/1748-9326/ac8754.

Zhen Qu, Daniel J. Jacob, A. Anthony Bloom, John R. Worden, Robert J. Parker, and Hartmut Boesch. Inverse modeling of 2010–2022 satellite observations shows that inundation of the wet tropics drove the 2020–2022 methane surge. *Proceedings of the National Academy of Sciences*, 121(40):e2402730121, October 2024. doi: 10.1073/pnas.2402730121.

David S. Stevenson, Richard G. Derwent, Oliver Wild, and William J. Collins. COVID-19 lockdown emission reductions have the potential to explain over half of the coincident increase in global atmospheric methane. *Atmospheric Chemistry and Physics*, 22(21):14243–14252, November 2022. ISSN 1680-7316. doi: 10.5194/acp-22-14243-2022.